GOVIND PATHAK

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PROFESSIONAL SUMMARY

MSc Artificial Intelligence & Machine Learning graduate (First Class Honours) from the University of Limerick. Built end-to-end ML projects in Python spanning federated learning, reinforcement learning, computer vision and AWS deployment (SageMaker, Lambda, API Gateway, RDS, Bedrock). Seeking graduate applied ML / data science roles.

TECHNICAL SKILLS

Programming Languages: Python (primary), SQL, R, C#
ML / Deep Learning: CNNs, DQN/DDQN, Federated Learning, Bayesian Logistic Regression, MC Dropout, XGBoost
Cloud & MLOps: AWS (SageMaker, Lambda, EC2, API Gateway, RDS, Bedrock), Git, Jupyter
Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Optuna, DEAP, Pandas, NumPy
NLP / Computer Vision: Faster R-CNN, TF-IDF, LDA, embeddings
Tools: Tableau, Excel (Power Query, pivot tables), Unity (C#)

EDUCATION

MSc Artificial Intelligence and Machine Learning Sep 2025 – Aug 2026
University of Limerick, Ireland | First Class Honours | QCA 3.45 / 4.0

B.E. Computer Science & Engineering (Artificial Intelligence & Machine Learning) Aug 2021 – Jun 2025
University of Mumbai, India | First Class Honours | CGPA 8.70 / 10

PROJECTS

Mental Health Risk Prediction – AWS AI Ecosystem Apr 2026
Python, AWS SageMaker, Lambda, API Gateway, Bedrock, RDS, EC2, Flask

- Developed and deployed a TF-IDF + Logistic Regression classifier (GoEmotions dataset) as a real-time SageMaker endpoint, served through a Flask web UI on EC2.
- Built the serverless backend with Lambda, API Gateway and RDS MySQL, and integrated Amazon Bedrock to generate plain-language explanations for each prediction, across six AWS services.

Probabilistic Federated Learning Approach for Mental Health Prediction (MSc Dissertation) Mar 2026 – Aug 2026
Python, PyTorch, Optuna, NHANES

- Built a federated learning pipeline across 4 clinical silos (obesity, hypertension, diabetes, general) using 36,259 NHANES participants to predict depression via PHQ-9.
- Implemented Bayesian Logistic Regression with Laplace-approximated per-silo uncertainty and precision-weighted aggregation, compared against FedAvg, plus an MC Dropout MLP for predictive uncertainty. Tuned models with Optuna (TPE).
- Precision-weighted aggregation matched FedAvg (AUROC 0.699 vs 0.701; balanced accuracy 0.644 vs 0.644) while adding per-silo uncertainty estimates.

Bone Fracture Detection (group project)
Faster R-CNN, ResNet-50 FPN

- Trained a Faster R-CNN model with a ResNet-50 FPN backbone to detect bone fractures, reaching mAP@0.5 of 0.6294.

DQN and Double DQN Agent – Atari Atlantis Apr 2026
Python, TensorFlow, Keras, Gymnasium, ALE

- Led a two-person study comparing Vanilla DQN and Double DQN over 1,800 episodes on ALE/Atlantis-v5; DDQN's 100-episode average score was 14.2% higher.
- Replaced a list-based replay buffer with a pre-allocated NumPy circular buffer, fixing GPU under-utilisation and cutting memory use.

EXPERIENCE

Education Support Worker (Academic Notetaker) | Servisource at TUS Moylish Sep 2026 – Present

- Support a student across their full timetable, about 20 hours a week.

Data Science & Data Analysis Intern | Cognifyz Technologies | Remote, Mumbai, India Jun 2024 – Aug 2024

- Performed data exploration, missing value treatment, class imbalance analysis and statistical profiling on a restaurant dataset.
- Engineered predictive features (affordability index, cuisine count, service flags) and ran geospatial analysis on latitude/longitude data.

Private Tuition Teacher & Academic Mentor | Wisdom Classes | Mumbai, India Jun 2020 – Aug 2025

- Tutored Maths and Science for Grades 8–12, preparing students for board exams with structured study plans and targeted practice.

CERTIFICATIONS & LANGUAGES

Epigeum Research Integrity (2026) | AWS Academy Cloud Foundations & Machine Learning Foundations (2026) | IBM AI Fundamentals (Dec 2025) | NPTEL Natural Language Processing (IIT Kharagpur) | English – Fluent